

# Niraj K. Jha

professor

## 基本信息

姓名: Niraj K. Jha                      所属院校: 普林斯顿大学  
邮箱: jha@princeton.edu              地区: United States

## 学科领域

人工智能    电子与计算机工程    集成电路设计与集成系统

## 研究领域

深度学习与神经网络    低功耗设计与能耗优化    计算机体系结构    数字集成电路设计  
EDA工具与电路验证    智能边缘计算    集成电路设计与制造    存储器设计与架构  
低功耗电路与能量管理    微电子器件与制造    +2

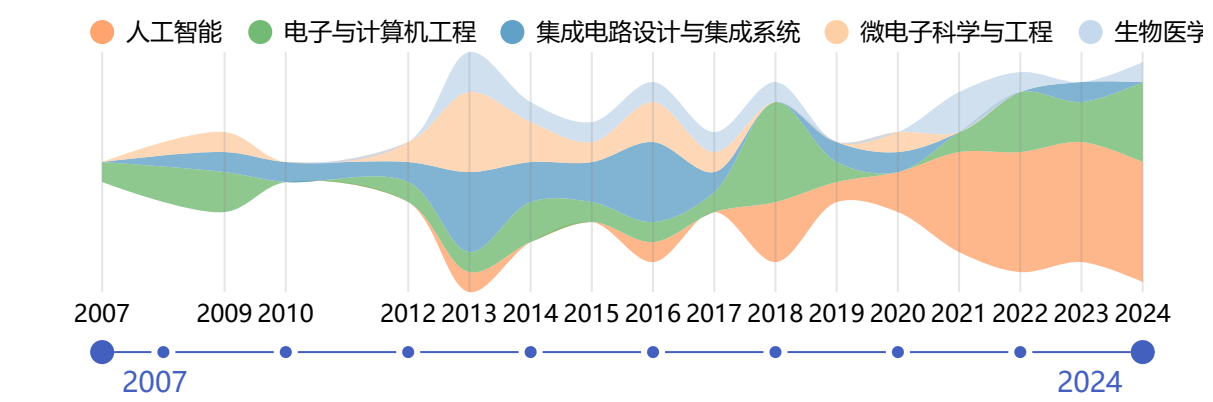
## 个人简介

Niraj K. Jha is a distinguished Professor of Electrical Engineering at Princeton University, located in Princeton, NJ. His research focuses on smart healthcare, transformer synthesis, transformer-accelerator co-design, Internet-of-Things, counterfactual reasoning, efficient randomized controlled trials, and artificial general intelligence. Professor Jha can be reached at his office B-220, E-QUAD, via phone at (609) 258-4754, fax at (609) 258-3745, or email at jha@princeton.edu.

## 工作经历

Princeton University - Professor

近年发表的论文



TAD-SIE: sample size estimation for clinical randomized controlled trials using a Trend-Adaptive Design with a Synthetic-Intervention-Based Estimator

TRIALS

Lala, Sayeri; Jha, Niraj K.

被引用次数: 0 2025

EdgeTran: Device-Aware Co-Search of Transformers for Efficient Inference on Mobile Edge Platforms

IEEE TRANSACTIONS ON MOBILE COMPUTING

Tuli, Shikhar; Jha, Niraj K.

被引用次数: 0 2024

Attention-Driven Training-Free Efficiency Enhancement of Diffusion Models

2024 IEEE/CVF CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION (CVPR)

Wang, Hongjie; Liu, Difan; Kang, Yan; Lie, Yijun; Line, Zhe; Jha, Niraj K.; Liu, Yuchen

被引用次数: 0 2024

Zero-TPPrune: Zero-Shot Token Pruning through Leveraging of the Attention Graph in Pre-Trained Transformers

2024 IEEE/CVF CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION (CVPR)

Wang, Hongjie; Dedhia, Bhishma; Jha, Niraj K.

被引用次数: 2 2024

DOCTOR: A Multi-Disease Detection Continual Learning Framework Based on Wearable Medical Sensors

ACM TRANSACTIONS ON EMBEDDED COMPUTING SYSTEMS

Li, Chia-Hao; Jha, Niraj K.

被引用次数: 0 2024

SECRETS: Subject-efficient clinical randomized controlled trials using synthetic intervention

CONTEMPORARY CLINICAL TRIALS COMMUNICATIONS

Lala, Sayeri; Jha, Niraj K.

|                                                                                                                                                                                    |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 被引用次数: 1                                                                                                                                                                           | 2024 |
| <hr/>                                                                                                                                                                              |      |
| <b>Attention-Driven Training-Free Efficiency Enhancement of Diffusion Models</b>                                                                                                   |      |
| IEEE/CVF Computer Vision and Pattern Recognition Conference                                                                                                                        |      |
| Hongjie Wang; Difan Liu; Yan Kang; Yijun Li; Zhe Lin; Niraj K. Jha; Yuchen Liu                                                                                                     |      |
| 被引用次数: 0                                                                                                                                                                           | 2024 |
| <hr/>                                                                                                                                                                              |      |
| <b>Zero-TPrune: Zero-Shot Token Pruning Through Leveraging of the Attention Graph in Pre-Trained Transformers</b>                                                                  |      |
| IEEE/CVF Computer Vision and Pattern Recognition Conference                                                                                                                        |      |
| Hongjie Wang; Bhishma Dedhia; Niraj K. Jha                                                                                                                                         |      |
| 被引用次数: 0                                                                                                                                                                           | 2024 |
| <hr/>                                                                                                                                                                              |      |
| <b>TransCODE: Co-Design of Transformers and Accelerators for Efficient Training and Inference</b>                                                                                  |      |
| IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS                                                                                                      |      |
| Tuli, Shikhar; Jha, Niraj K.                                                                                                                                                       |      |
| 被引用次数: 2                                                                                                                                                                           | 2023 |
| <hr/>                                                                                                                                                                              |      |
| <b>INFORM: Inverse Design Methodology for Constrained Multiobjective Optimization</b>                                                                                              |      |
| IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS                                                                                                      |      |
| Terway, Prerit; Jha, Niraj K.                                                                                                                                                      |      |
| 被引用次数: 2                                                                                                                                                                           | 2023 |
| <hr/>                                                                                                                                                                              |      |
| <b>TUTOR: Training Neural Networks Using Decision Rules as Model Priors</b>                                                                                                        |      |
| IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS                                                                                                      |      |
| Hassantabar, Shayan; Terway, Prerit; Jha, Niraj K.                                                                                                                                 |      |
| 被引用次数: 1                                                                                                                                                                           | 2023 |
| <hr/>                                                                                                                                                                              |      |
| <b>CODEBench: A Neural Architecture and Hardware Accelerator Co-Design Framework</b>                                                                                               |      |
| ACM TRANSACTIONS ON EMBEDDED COMPUTING SYSTEMS                                                                                                                                     |      |
| Tuli, Shikhar; Li, Chia-Hao; Sharma, Ritvik; Jha, Niraj K.                                                                                                                         |      |
| 被引用次数: 4                                                                                                                                                                           | 2023 |
| <hr/>                                                                                                                                                                              |      |
| <b>How MagNet: Machine Learning Framework for Modeling Power Magnetic Material Characteristics</b>                                                                                 |      |
| IEEE TRANSACTIONS ON POWER ELECTRONICS                                                                                                                                             |      |
| Li, Haoran; Serrano, Diego; Guillod, Thomas; Wang, Shukai; Dogariu, Evan; Nadler, Andrew; Luo, Min; Bansal, Vineet; Jha, Niraj K.; Chen, Yuxin; Sullivan, Charles R.; Chen, Minjie |      |
| 被引用次数: 17                                                                                                                                                                          | 2023 |
| <hr/>                                                                                                                                                                              |      |
| <b>Why MagNet: Quantifying the Complexity of Modeling Power Magnetic Material Characteristics</b>                                                                                  |      |
| IEEE TRANSACTIONS ON POWER ELECTRONICS                                                                                                                                             |      |
| Serrano, Diego; Li, Haoran; Wang, Shukai; Guillod, Thomas; Luo, Min; Bansal, Vineet; Jha, Niraj K.; Chen, Yuxin; Sullivan, Charles R.; Chen, Minjie                                |      |

被引用次数: 26

2023

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### **Im-Promptu: In-Context Composition from Image Prompts**

ADVANCES IN NEURAL INFORMATION PROCESSING SYSTEMS 36 (NEURIPS 2023)

Dedhia, Bhishma; Chang, Michael; Snell, Jake C.; Griffiths, Thomas L.; Jha, Niraj K.

被引用次数: 0

2023

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### **REPAIRS: Gaussian Mixture Model-based Completion and Optimization of Partially Specified Systems**

ACM TRANSACTIONS ON EMBEDDED COMPUTING SYSTEMS

Terway, Prerit; Jha, Niraj K.

被引用次数: 1

2023

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### **Fast Design Space Exploration of Nonlinear Systems: Part II**

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Terway, Prerit; Hamidouche, Kenza; Jha, Niraj K.

被引用次数: 4

2022

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### **Machine Learning Assisted Security Analysis of 5G-Network-Connected Systems**

IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING

Saha, Tanujay; Aaraj, Najwa; Jha, Niraj K.

被引用次数: 12

2022

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### **GRAVITAS: Graphical Reticulated Attack Vectors for Internet-of-Things Aggregate Security**

IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING

Brown, Jacob; Saha, Tanujay; Jha, Niraj K.

被引用次数: 7

2022

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### **MHDeep: Mental Health Disorder Detection System Based on Wearable Sensors and Artificial Neural Networks**

ACM TRANSACTIONS ON EMBEDDED COMPUTING SYSTEMS

Hassantabar, Shayan; Zhang, Joe; Yin, Hongxu; Jha, Niraj K.

被引用次数: 13

2022

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### **SCANN: Synthesis of Compact and Accurate Neural Networks**

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Hassantabar, Shayan; Wang, Zeyu; Jha, Niraj K.

被引用次数: 11

2022

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### **Towards Execution-Efficient LSTMs via Hardware-Guided Grow-and-Prune Paradigm**

IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING

Yin, Hongxu; Chen, Guoyang; Li, Yingmin; Che, Shuai; Zhang, Weifeng; Jha, Niraj K.

被引用次数: 0

2022

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### **CURIOUS: Efficient Neural Architecture Search Based on a Performance Predictor and Evolutionary Search**

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Hassantabar, Shayan; Dai, Xiaoliang; Jha, Niraj K.

被引用次数: 3

2022

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### ML-FEED: Machine Learning Framework for Efficient Exploit Detection

2022 IEEE 4TH INTERNATIONAL CONFERENCE ON TRUST, PRIVACY AND SECURITY IN INTELLIGENT SYSTEMS, AND APPLICATIONS, TPS-ISA

Saha, Tanujay; Al Rahat, Tamjid; Aaraj, Najwa; Tian, Yuan; Jha, Niraj K.

被引用次数: 0

2022

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### SECRET: Semantically Enhanced Classification of Real-World Tasks

IEEE TRANSACTIONS ON COMPUTERS

Akmandor, Ayten Ozge; Ortiz, Jorge; Manotas, Irene; Ko, Bongjun; Jha, Niraj K.

被引用次数: 1

2021

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### DiabDeep: Pervasive Diabetes Diagnosis Based on Wearable Medical Sensors and Efficient Neural Networks

IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING

Yin, Hongxu; Mukadam, Bilal; Dai, Xiaoliang; Jha, Niraj K.

被引用次数: 24

2021

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### CovidDeep: SARS-CoV-2/COVID-19 Test Based on Wearable Medical Sensors and Efficient Neural Networks

IEEE TRANSACTIONS ON CONSUMER ELECTRONICS

Hassantabar, Shayan; Stefano, Novati; Ghanakota, Vishweshwar; Ferrari, Alessandra; Nicola, Gregory N.; Bruno, Raffaele; Marino, Ignazio R.; Hamidouche, Kenza; Jha, Niraj K.

被引用次数: 35

2021

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### YSUY: Your Smartphone Understands You-Using Machine Learning to Address Fundamental Human Needs

IEEE TRANSACTIONS ON SYSTEMS MAN CYBERNETICS-SYSTEMS

Akmandor, Ayten Ozge; Dai, Xiaoliang; Jha, Niraj K.

被引用次数: 4

2021

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### Software-Defined Design Space Exploration for an Efficient DNN Accelerator Architecture

IEEE TRANSACTIONS ON COMPUTERS

Yu, Ye; Li, Yingmin; Che, Shuai; Jha, Niraj K.; Zhang, Weifeng

被引用次数: 14

2021

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### McPAT-Monolithic: An Area/Power/Timing Architecture Modeling Framework for 3-D Hybrid Monolithic Multicore Systems

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Guler, Abdullah; Jha, Niraj K.

被引用次数: 15

2020

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### INVITED: Efficient Synthesis of Compact Deep Neural Networks

Design Automation Conference

Wenhan Xia; Hongxu Yin; Niraj K. Jha

|                                                                                                                                   |      |
|-----------------------------------------------------------------------------------------------------------------------------------|------|
| 被引用次数: 0                                                                                                                          | 2020 |
| <hr/>                                                                                                                             |      |
| <b>Grow and Prune Compact, Fast, and Accurate LSTMs</b>                                                                           |      |
| IEEE TRANSACTIONS ON COMPUTERS                                                                                                    |      |
| Dai, Xiaoliang; Yin, Hongxu; Jha, Niraj K.                                                                                        |      |
| 被引用次数: 55                                                                                                                         | 2020 |
| <hr/>                                                                                                                             |      |
| <b>Three-Dimensional Monolithic FinFET-Based 8T SRAM Cell Design for Enhanced Read Time and Low Leakage</b>                       |      |
| IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS                                                                  |      |
| Guler, Abdullah; Jha, Niraj K.                                                                                                    |      |
| 被引用次数: 18                                                                                                                         | 2019 |
| <hr/>                                                                                                                             |      |
| <b>NeST: A Neural Network Synthesis Tool Based on a Grow-and-Prune Paradigm</b>                                                   |      |
| IEEE TRANSACTIONS ON COMPUTERS                                                                                                    |      |
| Dai, Xiaoliang; Yin, Hongxu; Jha, Niraj K.                                                                                        |      |
| 被引用次数: 138                                                                                                                        | 2019 |
| <hr/>                                                                                                                             |      |
| <b>Detecting and Alleviating Stress with SoDA</b>                                                                                 |      |
| COMPUTER                                                                                                                          |      |
| Akmandor, Ayten Ozge; Jha, Niraj K.                                                                                               |      |
| 被引用次数: 1                                                                                                                          | 2018 |
| <hr/>                                                                                                                             |      |
| <b>Energy-Efficient Monolithic Three-Dimensional On-Chip Memory Architectures</b>                                                 |      |
| IEEE TRANSACTIONS ON NANOTECHNOLOGY                                                                                               |      |
| Yu, Ye; Jha, Niraj K.                                                                                                             |      |
| 被引用次数: 20                                                                                                                         | 2018 |
| <hr/>                                                                                                                             |      |
| <b>Genetic Programming for Energy-Efficient and Energy-Scalable Approximate Feature Computation in Embedded Inference Systems</b> |      |
| IEEE TRANSACTIONS ON COMPUTERS                                                                                                    |      |
| Lu, Jie; Jia, Hongyang; Verma, Naveen; Jha, Niraj K.                                                                              |      |
| 被引用次数: 10                                                                                                                         | 2018 |
| <hr/>                                                                                                                             |      |
| <b>Hybrid Monolithic 3-D IC Floorplanner</b>                                                                                      |      |
| IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS                                                                  |      |
| Guler, Abdullah; Jha, Niraj K.                                                                                                    |      |
| 被引用次数: 12                                                                                                                         | 2018 |
| <hr/>                                                                                                                             |      |
| <b>Smart Health Care An edge-side computing perspective</b>                                                                       |      |
| IEEE CONSUMER ELECTRONICS MAGAZINE                                                                                                |      |
| Akmandor, Ayten Ozge; Jha, Niraj K.                                                                                               |      |
| 被引用次数: 42                                                                                                                         | 2018 |
| <hr/>                                                                                                                             |      |
| <b>Statistical Optimization of FinFET Processor Architectures under PVT Variations Using Dual Device-Type Assignment</b>          |      |
| ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS                                                                         |      |

Yu, Ye; Jha, Niraj K.

被引用次数: 2

2018

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**Automated Quantum Circuit Synthesis and Cost Estimation for the Binary Welded Tree Oracle**

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Ghosh, Mrityunjay; Chakrabarti, Amlan; Jha, Niraj K.

被引用次数: 6

2017

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**Using a Device State Library to Boost the Performance of TCAD Mixed-Mode Simulation**

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Dai, Xiaoliang; Jha, Niraj K.

被引用次数: 1

2017

---

**Improving Convergence and Simulation Time of Quantum Hydrodynamic Simulation: Application to Extraction of Best 10-nm FinFET Parameter Values**

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Dai, Xiaoliang; Jha, Niraj K.

被引用次数: 5

2017

---

**Ultra-low-leakage, Robust FinFET SRAM Design Using Multiparameter Asymmetric FinFETs**

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Guler, Abdullah; Jha, Niraj K.

被引用次数: 4

2017

---

**CABA: Continuous Authentication Based on BioAura**

IEEE TRANSACTIONS ON COMPUTERS

Mosenia, Arsalan; Sur-Kolay, Susmita; Raghunathan, Anand; Jha, Niraj K.

被引用次数: 51

2017

---

**Ultra-Low-Leakage and High-Performance Logic Circuit Design Using Multiparameter Asymmetric FinFETs**

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Chaudhuri, Sourindra M.; Jha, Niraj K.

被引用次数: 2

2016

---

**Delay/Power Modeling and Optimization of FinFET Circuit Modules under PVT Variations: Observing the Trends between the 22nm and 14nm Technology Nodes**

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Tang, Aoxiang; Gao, Xun; Chen, Lung-Yen; Jha, Niraj K.

被引用次数: 4

2016

---

**TCAD-Assisted Capacitance Extraction of FinFET SRAM and Logic Arrays**

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Bhattacharya, Debajit; Jha, Niraj K.

被引用次数: 4

2016

---

**GenFin: Genetic Algorithm-Based Multiobjective Statistical Logic Circuit Optimization Using Incremental Statistical Analysis**

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Tang, Aoxiang; Jha, Niraj K.

被引用次数: 10

2016

**A 3-D CPU-FPGA-DRAM Hybrid Architecture for Low-Power Computation**

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chen, Xianmin; Jha, Niraj K.

被引用次数: 10

2016

**Compressed Signal Processing on Nyquist-Sampled Signals**

IEEE TRANSACTIONS ON COMPUTERS

Lu, Jie; Verma, Naveen; Jha, Niraj K.

被引用次数: 12

2016

**Ultra-High Density Monolithic 3-D FinFET SRAM With Enhanced Read Stability**

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Bhattacharya, Debajit; Jha, Niraj K.

被引用次数: 17

2016

**Signal Processing With Direct Computations on Compressively Sensed Data**

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Shoaib, Mohammed; Jha, Niraj K.; Verma, Naveen

被引用次数: 39

2015

**FDR 2.0: A Low-Power Dynamically Reconfigurable Architecture and Its FinFET Implementation**

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lin, Ting-Jung; Zhang, Wei; Jha, Niraj K.

被引用次数: 6

2015

**McPAT-PVT: Delay and Power Modeling Framework for FinFET Processor Architectures Under PVT Variations**

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Tang, Aoxiang; Yang, Yang; Lee, Chun-Yi; Jha, Niraj K.

被引用次数: 30

2015

**PAQCS: Physical Design-Aware Fault-Tolerant Quantum Circuit Synthesis**

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lin, Chia-Chun; Sur-Kolay, Susmita; Jha, Niraj K.

被引用次数: 44

2015

**Design of Efficient Content Addressable Memories in High-Performance FinFET Technology**

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS



Bhattacharya, Debajit; Bhoj, Ajay N.; Jha, Niraj K.

被引用次数: 10

2015

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### Systematic Poisoning Attacks on and Defenses for Machine Learning in Healthcare

IEEE JOURNAL OF BIOMEDICAL AND HEALTH INFORMATICS

Mozaffari-Kermani, Mehran; Sur-Kolay, Susmita; Raghunathan, Anand; Jha, Niraj K.

被引用次数: 172

2015

---

### A defense framework against malware and vulnerability exploits

INTERNATIONAL JOURNAL OF INFORMATION SECURITY

Zhang, Meng; Raghunathan, Anand; Jha, Niraj K.

被引用次数: 13

2014

---

### FinPrin: FinFET Logic Circuit Analysis and Optimization Under PVT Variations

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Yang, Yang; Jha, Niraj K.

被引用次数: 28

2014

---

### A Fine-Grain Dynamically Reconfigurable Architecture Aimed at Reducing the FPGA-ASIC Gaps

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lin, Ting-Jung; Zhang, Wei; Jha, Niraj K.

被引用次数: 12

2014

---

### FTQLS: Fault-Tolerant Quantum Logic Synthesis

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lin, Chia-Chun; Chakrabarti, Amlan; Jha, Niraj K.

被引用次数: 21

2014

---

### FinCANON: A PVT-Aware Integrated Delay and Power Modeling Framework for FinFET-Based Caches and On-Chip Networks

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lee, Chun-Yi; Jha, Niraj K.

被引用次数: 21

2014

---

### Parasitics-Aware Design of Symmetric and Asymmetric Gate-Workfunction FinFET SRAMs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Bhoj, Ajay N.; Jha, Niraj K.

被引用次数: 23

2014

---

### Trustworthiness of Medical Devices and Body Area Networks

PROCEEDINGS OF THE IEEE

Zhang, Meng; Raghunathan, Anand; Jha, Niraj K.

被引用次数: 91

2014

---

### 3D vs. 2D Device Simulation of FinFET Logic Gates under PVT Variations

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Chaudhuri, Sourindra M.; Jha, Niraj K.

被引用次数: 2

2014

---

**MedMon: Securing Medical Devices Through Wireless Monitoring and Anomaly Detection**

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Zhang, Meng; Raghunathan, Anand; Jha, Niraj K.

被引用次数: 109

2013

---

**Thermal Characterization of Test Techniques for FinFET and 3D Integrated Circuits**

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Tang, Aoxiang; Jha, Niraj K.

被引用次数: 2

2013

---

**Design of Logic Gates and Flip-Flops in High-performance FinFET Technology**

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Bhoj, Ajay N.; Jha, Niraj K.

被引用次数: 48

2013

---

**3-D-TCAD-Based Parasitic Capacitance Extraction for Emerging Multigate Devices and Circuits**

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Bhoj, Ajay N.; Joshi, Rajiv V.; Jha, Niraj K.

被引用次数: 30

2013

---

**Optimized Quantum Gate Library for Various Physical Machine Descriptions**

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lin, Chia-Chun; Chakrabarti, Amlan; Jha, Niraj K.

被引用次数: 22

2013

---

**Algorithm-Driven Architectural Design Space Exploration of Domain-Specific Medical-Sensor Processors**

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Shoaib, Mohammed; Jha, Niraj K.; Verma, Naveen

被引用次数: 7

2013

---

**Variable-Pipeline-Stage Router**

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lee, Chun-Yi; Jha, Niraj K.

被引用次数: 9

2013

---

**Efficient Methodologies for 3-D TCAD Modeling of Emerging Devices and Circuits**

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Bhoj, Ajay N.; Joshi, Rajiv V.; Jha, Niraj K.

被引用次数: 10

2013

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**Fault Models for Logic Circuits in the Multigate Era**

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Bhoj, Ajay N.; Simsir, Muzaffer O.; Jha, Niraj K.

被引用次数: 24

2012

---

### Secure Reconfiguration of Software-Defined Radio

ACM TRANSACTIONS ON EMBEDDED COMPUTING SYSTEMS

Li, Chunxiao; Jha, Niraj K.; Raghunathan, Anand

被引用次数: 5

2012

---

### INVISIOS: A Lightweight, Minimally Intrusive Secure Execution Environment

ACM TRANSACTIONS ON EMBEDDED COMPUTING SYSTEMS

Arora, Divya; Aaraj, Najwa; Raghunathan, Anand; Jha, Niraj K.

被引用次数: 1

2012

---

### SRAM-Based NATURE: A Dynamically Reconfigurable FPGA Based on 10T Low-Power SRAMs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lin, Ting-Jung; Zhang, Wei; Jha, Niraj K.

被引用次数: 13

2012

---

### FinFET-Based Power Management for Improved DPA Resistance with Low Overhead

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Zhang, Meng; Jha, Niraj K.

被引用次数: 14

2011

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### FinFET-Based Power Simulator for Interconnection Networks

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Lee, Chun-Yi; Jha, Niraj K.

被引用次数: 5

2010

---

### Gated-Diode FinFET DRAMs: Device and Circuit Design-Considerations

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Bhoj, Ajay N.; Jha, Niraj K.

被引用次数: 10

2010

---

### A Hybrid Nano-CMOS Architecture for Defect and Fault Tolerance

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Simsir, Muzaffer O.; Cadambi, Srihari; Ivancic, Franjo; Roetteler, Martin; Jha, Niraj K.

被引用次数: 7

2009

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### Fast Enhancement of Validation Test Sets for Improving the Stuck-at Fault Coverage of RTL Circuits

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lingappan, Loganathan; Gangaram, Vijay; Jha, Niraj K.; Chakravarty, Sreejit

被引用次数: 1

2009

---

### A Hybrid Nano/CMOS Dynamically Reconfigurable System-Part II: Design Optimization Flow

**An energy-aware framework for dynamic software management in mobile computing systems**

ACM TRANSACTIONS ON EMBEDDED COMPUTING SYSTEMS

Fei, Yunsu; Zhong, Lin; Jha, Niraj K.

被引用次数: 14

2008

---

**Energy-optimizing source code transformations for operating system-driven embedded software**

ACM TRANSACTIONS ON EMBEDDED COMPUTING SYSTEMS

Fei, Yunsu; Ravi, Srivaths; Raghunathan, Anand; Jha, Niraj K.

被引用次数: 16

2008

---

**Hybrid simulation for energy estimation of embedded software**

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Muttreja, Anish; Raghunathan, Anand; Ravi, Srivaths; Jha, Niraj K.

被引用次数: 6

2007

---

**CURIOUS: Efficient Neural Architecture Search Based on a Performance Predictor and Evolutionary Search**

TCAD

Shayan Hassantabar; Xiaoliang Dai; Niraj K. Jha

被引用次数: 0

---

**EdgeTran: Device-Aware Co-Search of Transformers for Efficient Inference on Mobile Edge Platforms**

TMC

Shikhar Tuli; Niraj K. Jha

被引用次数: 0

---

**ChamNet: Towards Efficient Network Design Through Platform-Aware Model Adaptation**

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Xiaoliang Dai; Peizhao Zhang; Bichen Wu; Hongxu Yin; Fei Sun; Yanghan Wang; Marat Dukhan; Yunqing Hu; Yiming Wu; Yangqing Jia; Peter Vajda; Matt Uyttendaele; Niraj K. Jha

被引用次数: 0

---

**TUTOR: Training Neural Networks Using Decision Rules as Model Priors**

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Shayan Hassantabar; Prerit Terway; Niraj K. Jha

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---

**SCANN: Synthesis of Compact and Accurate Neural Networks**

TCAD

Shayan Hassantabar; Zeyu Wang; Niraj K. Jha

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---

### AccelTran: A Sparsity-Aware Accelerator for Dynamic Inference With Transformers

TCAD

Shikhar Tuli; Niraj K. Jha

被引用次数: 0

---

### TransCODE: Co-Design of Transformers and Accelerators for Efficient Training and Inference

TCAD

Shikhar Tuli; Niraj K. Jha

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---

### NeST: A Neural Network Synthesis Tool Based on a Grow-and-Prune Paradigm

TC

Xiaoliang Dai; Hongxu Yin; Niraj K. Jha

被引用次数: 0

---

### Grow and Prune Compact, Fast, and Accurate LSTMs

TC

Xiaoliang Dai; Hongxu Yin; Niraj K. Jha

被引用次数: 0

---

### CABA: Continuous Authentication Based on BioAura

TC

Arsalan Mosenia; Susmita Sur-Kolay; Anand Raghunathan; Niraj K. Jha

被引用次数: 0

---

### Software-Defined Design Space Exploration for an Efficient DNN Accelerator Architecture

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Ye Yu; Yingmin Li; Shuai Che; Niraj K. Jha; Weifeng Zhang

被引用次数: 0

---

### Genetic Programming for Energy-Efficient and Energy-Scalable Approximate Feature Computation in Embedded Inference Systems

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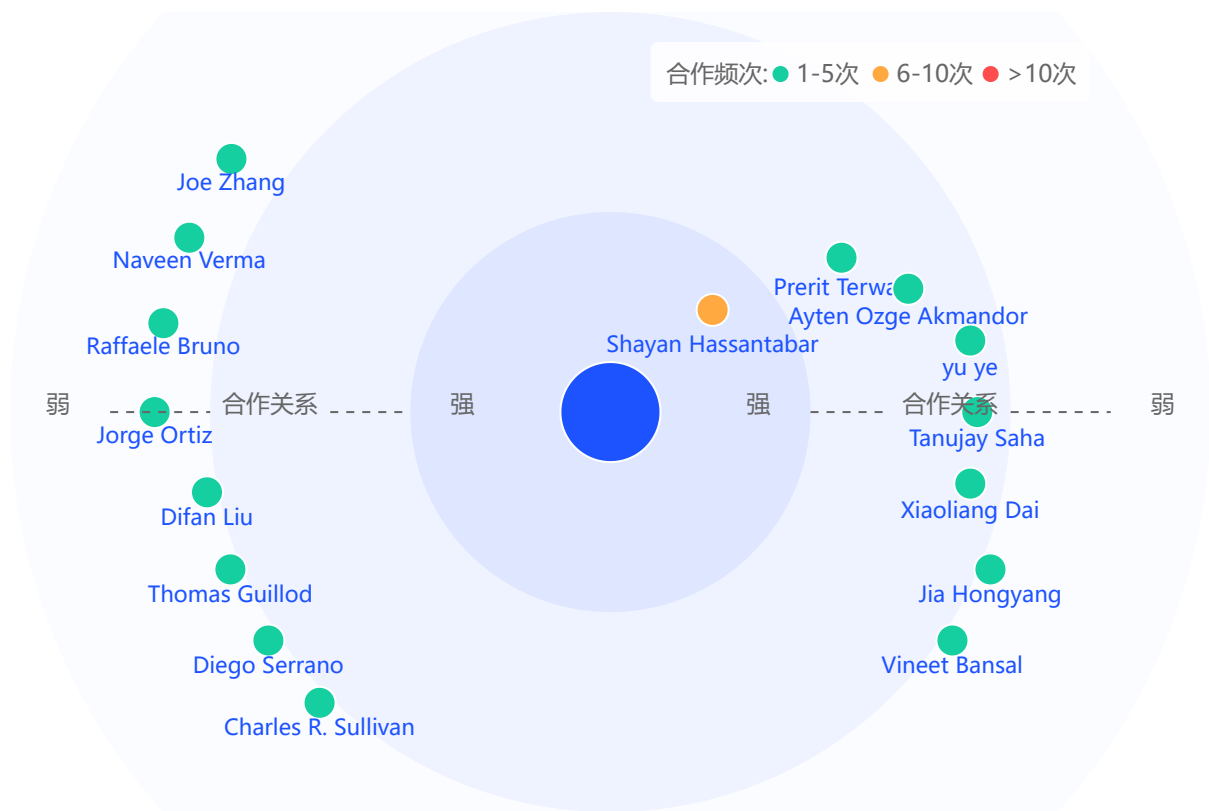
Jie Lu; Hongyang Jia; Naveen Verma; Niraj K. Jha

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## 学术合作

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